

DINGCHENG YI

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EDUCATION

Bachelor of Engineering in Bioinformatics, Peking University

Sep. 2021 - Jul. 2025

- **Shen Tong Outstanding Undergraduate Award**, highest honor for undergraduates in the department
- **May 4th Scholarship**

Ph.D. in Systems Biology, Columbia University

Sep. 2025 - Present

PUBLICATIONS

- Chao Zhang*, Yuxuan Sun*, **Dingcheng Yi***, Benyuan Jiang*, Lixu Yan*, ..., Zemin Zhang, Wenzhao Zhong. Neoadjuvant sintilimab plus chemotherapy in early-stage EGFR-mutant NSCLC: phase 2 trial interim results (NEOTIDE/CTONG2104). *Cell Reports Medicine*, **2024**
- Zedao Liu*, Zhenlin Yang*, Junqi Wu*, Wenjie Zhang*, Yuxuan Sun*, Chao Zhang*, Guangyu Bai*, ..., **Dingcheng Yi**, ..., Zemin Zhang A single-cell atlas reveals immune heterogeneity in anti-PD-1-treated non-small cell lung cancer. *Cell*, **2025**
- Wen Si*, Sijin Cheng*, Haiyin He*, Yu Zhang, Yuhui Miao, **Dingcheng Yi**, Mengjiao Ni, Anqiang Wang, Hongtao Fan, Yufei Bo, Chang Liu, Zhaode Bu, Linnan Zhu, Zemin Zhang SARDH in the 1-C metabolism sculpts the T-cell fate and serves as a potential cancer therapeutic target. *Cell Mol Immunol*, **2025**.

RESEARCH EXPERIENCE

Dr. David Knowles' Group, Columbia University

Xeno: Sequence-to-function modeling through latent chromatin state

Apr. 2026 - Present

- Developed a sequence-to-function model to predict celltype-specific omics profiles from DNA sequence with a latent chromatin state.
- Benchmarked the model against other SOTA methods(e.g. AlphaGenome) and figured out failure modes.
- Interpret the latent state within variant effect prediction task.

Dr. Bianca Dumitrascu's Group, Columbia University

Benchmarking Gene Regulatory Network(GRN) Inference Algorithms in Developement

Jan. 2026 - Mar. 2026

- Implemented a pipeline for measuring GRN inference methods' performance in determining cell fates.
- Simulated prototype GRN's impact on gene expression.
- Developed a mixture of Neural ODE model to learn the gene expression dynamics from developmental data.

Dr. Bing Ren's Group, Columbia University

Decoding Transcriptional Programs at Scale through Acute Protein Depletions

Sep. 2025 - Present

- Developed a pipeline for scSLAM-seq data preprocessing and increased recover rate of labeled mRNA molecule.
- Used a double demean strategy to distinguish true regulatory signal from experiment condition effects.
- Developed a variational inference framework for inferring the transcription dynamics
- Identified potential mRNA stability control mechanisms.

Dr. Xiaojie Qiu's Group, Stanford University

Deciphering Cell-Cell Communication in Mouse Heart Development using 3D MERFISH

Jun. 2024 - Aug. 2024

- Investigated cell-cell communication events and constructed a 3D CCC atlas in mouse heart development.
- Identified differential signal enrichment patterns in cardiac progenitor cells.
- Revealed Wnt and BMP pathway's interplay in juxtacardiac field development.
- Developed an explainable-AI model to infer CCC's impact on heart morphogenesis.

Dr. Zemin Zhang's Group, Peking University

Patient Phenotype Associated Spatial Community Feature Mining

Oct. 2023 - Jun. 2025

- Developed an efficient algorithm to identify co-located celltypes in spatial omics datasets.
- Demonstrated phenotype association and intepretability of detected spatial features.
- Leveraged statistical model to explore the relationships between spatial features and immunotherapy response.
- Investigated incorporating bulk RNA sequencing data with clinical information to guide the discovery of cell communities.

Single-Cell Analysis of EGFR-mutated NSCLC TME post Neoadjuvant Immunotherapy Sep. 2022 - Jul. 2024

- Systematically characterized the tumor microenvironment composition in EGFR-mutated lung cancer patients.
- Identified immune cell subtypes associated with patient treatment response.
- Proposed a patient stratification method using infiltration and clonal expansion patterns of two subtype of T cells and verified the reliability of stratification in multiple bulk RNA datasets.
- Co-authored a publication in *Cell Reports Medicine*, contributing to both manuscript writing and the rebuttal process.

Whole exome sequencing data analysis of Immunotherapy Response in NSCLC Patients Mar. 2022 - Dec. 2022

- Developed a pipeline for calling single nucleotide variants.
- Discovered lower tumor mutational burden among responders and attributed it to prior treatment.
- Found certain driver mutations associated with poor response and verified it through literature survey.

ACADEMIC EXPERIENCE

Undergraduate Honors Program in Biology(UHPB), Peking University Mar. 2023 - Jan. 2025

- Presented one and participated in at least four **journal clubs** every semester
 - **Main Host**: Multimodal Single Cell Data Integration
 - **Main Host**: Transforming Single Cell Biology: The Role of Transformer Models in Single Cell Omics
 - The Development and Clinical Application of T Cell Immunotherapy
 - Genomic Instability in Human Cancer
- Displayed a poster about the EGFR-mutant NSCLC project at the 10th Annual Symposium
- Presented the summer research project on CCI in mouse heart development at the 11th Annual Symposium

TEACHING EXPERIENCE

Teaching Assistant, Bioinformatics Laboratory, Peking University Mar. 2024 - Jun. 2024

- Evaluated students’ assignments and provided specific comments and suggestions for improvement.
- Explained complex analysis methods with biological intuition to facilitate understanding
- Fostered students’ problem-solving skills by encouraging them to think about each command’s outcome

HONORS & AWARDS

National Scholarship	2024
Qin Wanshun-Jin Yunhui Scholarship	2022
Award for Academic Excellence	2022, 2023
Pacemaker to Merit Student	2024
Shen Tong Outstanding Undergraduate Award, highest honor for undergraduates in the department	
May 4th Scholarship, highest individual scholarship awarded at Peking University	

SKILLS

Programming Skills

R, Python	Familiar with data analysis and visualization packages: ggplot2, dplyr, pandas, matplotlib Familiar with single cell and spatial omics packages: Seurat, Scanpy, Squidpy, Spateo
Linux	Familiar with machine learning/deep learning frameworks: scikit-learn, PyTorch, PyG Familiar with slurm for job scheduling and basic bash script Daily Arch Linux desktop user, familiar with linux system management